Unit: Unit 3: Functions
Topic: Inverse Relations and Functions

Name: ____________
Date: ________

Quiz (Carolina Reaper)

Q1. A shipping company uses the function $f(x) = 2x + 5$ to calculate delivery times. Which of the following is the inverse function?
(A) $f^{-1}(x) = \frac{x-5}{2}$
(B) $f^{-1}(x) = \frac{x+5}{2}$
(C) $f^{-1}(x) = 2x - 5$
(D) $f^{-1}(x) = \frac{x}{2} + 5$
(E) $f^{-1}(x) = \frac{x}{2} - 5$

Q2. If a function $f(x)$ has an inverse, then $f(x)$ is:
(A) always increasing
(B) always decreasing
(C) one-to-one
(D) periodic
(E) symmetric

Q3. The function $f(x) = x^2$ does not have an inverse because it is:
(A) not defined for all real numbers
(B) not continuous
(C) not one-to-one
(D) not onto
(E) not symmetric

Q4. A traffic flow model uses the function $f(x) = 3x - 2$ to calculate traffic volume. What is the inverse function?
(A) $f^{-1}(x) = \frac{x+2}{3}$
(B) $f^{-1}(x) = \frac{x-2}{3}$
(C) $f^{-1}(x) = 3x + 2$
(D) $f^{-1}(x) = \frac{x}{3} + 2$
(E) $f^{-1}(x) = \frac{x}{3} - 2$

Q5. To verify that $f(x) = 2x + 1$ and $g(x) = \frac{x-1}{2}$ are inverses, we need to show that:
(A) $f(g(x)) = x$ and $g(f(x)) = x$
(B) $f(g(x)) = g(f(x))$
(C) $f(x) = g(x)$
(D) $f(x) = g^{-1}(x)$
(E) $f(x) = f^{-1}(x)$

Q6. A logistics company uses the function $f(x) = x^3 - 2$ to calculate shipping costs. Which of the following is the inverse function?
(A) $f^{-1}(x) = \sqrt[3]{x+2}$
(B) $f^{-1}(x) = \sqrt[3]{x-2}$
(C) $f^{-1}(x) = x^3 + 2$
(D) $f^{-1}(x) = x^3 - 2$
(E) $f^{-1}(x) = \sqrt[3]{x} - 2$

Q7. The function $f(x) = e^x$ has an inverse because it is:
(A) one-to-one
(B) onto
(C) periodic
(D) symmetric
(E) not defined for all real numbers

Q8. A delivery service uses the function $f(x) = \frac{1}{x}$ to calculate delivery times. Which of the following is the inverse function?
(A) $f^{-1}(x) = \frac{1}{x}$
(B) $f^{-1}(x) = x$
(C) $f^{-1}(x) = \frac{1}{x^2}$
(D) $f^{-1}(x) = x^2$
(E) $f^{-1}(x) = \frac{1}{x} + 1$

Open Ended Questions (FRQ)

Q9. A transportation company uses the function $f(x) = 2x^2 + 3$ to calculate transportation costs. Find the inverse function and verify that it is indeed the inverse.

Q10. A shipping company uses the function $f(x) = x^3 - 2x + 1$ to calculate shipping costs. Determine if the function has an inverse and explain your reasoning.

Chef's Tips

- Tip 1: Encourage students to use algebraic methods to find inverse functions.
- Tip 2: Remind students to verify inverses using composition.
- Tip 3: Emphasize the importance of interpreting inverses in real-world contexts.